

4. Given the following C/C++ code below, suppose the CPU is executing the line `*result = sqrt(multiplier);` in the code. Draw the content of the active symbol tables at that point. Show the organization of the active symbol tables as a *single list of hash-tables* in your drawing and for each variable in a symbol table, write its name and type. Remember that, in C/C++ each `{ }` group represents a separate scope. That is, there are separate scopes for functions, for, and while loops. You do not need to separate the function header and body into two separate scopes, by the way. (Hint, when

you write a statement like `t n = v` in C, it means you are defining a new variable with the name `n` of the type `t` and assigning it an initial value `v`)

```
float a = 0.0;
float b = 0.0;
int table[10][20];

void main(int argc, char* argv[]) {
    a = atof(argv[1]);
    table[0][0] = atoi(argv[2]);

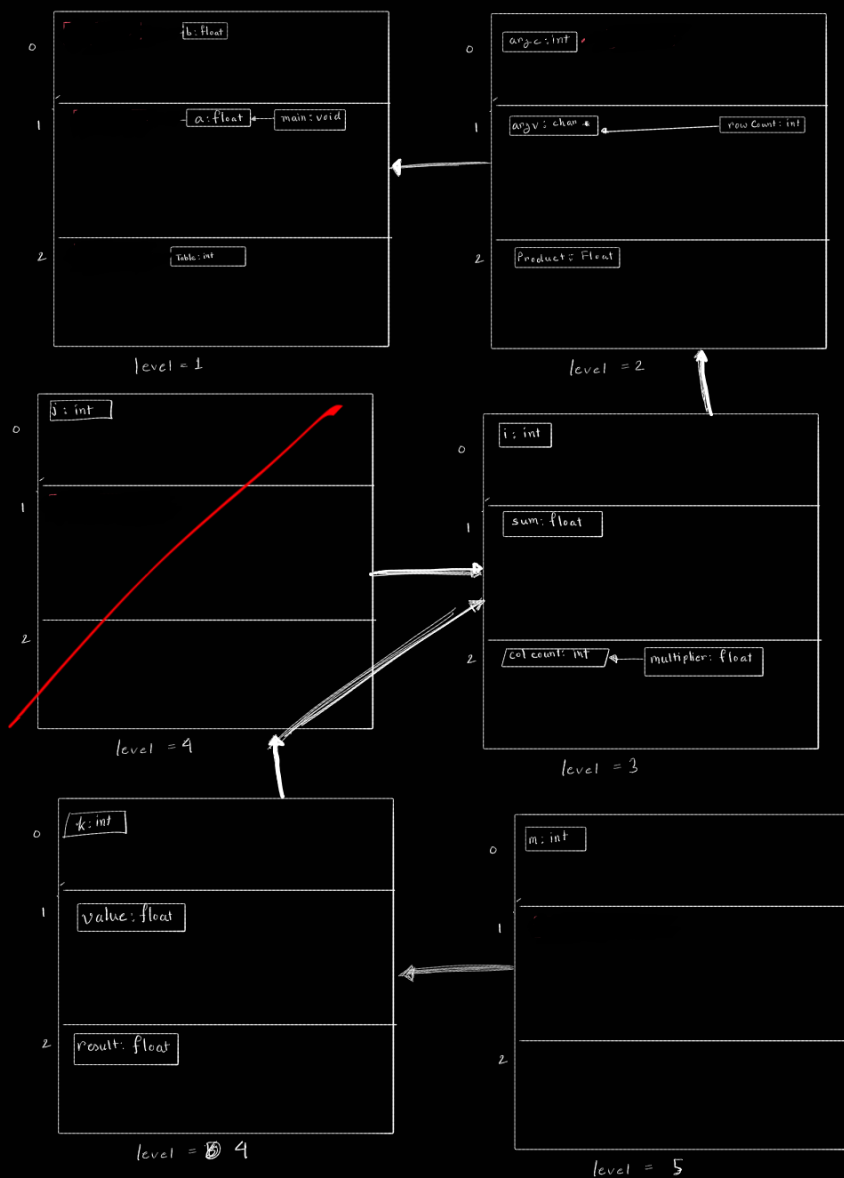
    float product = 1.0;
    int rowCount = 10;

    for (int i = 0; i < rowCount; i++) {
        float sum = 0.0;
        int colCount = 20;

        float multiplier = 1.0;
        for (int j = 0; j < 5; j++) {
            multiplier *= sqrt(a + j);
        }

        for (int k = 0; k < colCount; k++) {
            int value = table[i][k];
            float result = 1.0;

            for (int m = 0; m < value; m++) {
                result *= sqrt(multiplier);
                if (k > c) {
                    int c = 0;
                    c = x + k;
                }
                int l = 100;
            }
            sum += result;
        }
        product *= sum;
    }
}
```



Define Variables,

bucket_count = 3

hash function values

- `a = 1`
- `b = 0`
- `table = 2`
- `main = 1`
- `argc = 0`
- `argv = 1`
- `product = 2`
- `rowCount = 1`
- `i = 0`
- `sum = 1`
- `colCount = 2`
- `multiplier = 2`
- `j = 0`
- `k = 0`
- `value = 1`
- `result = 2`
- `m = 0`

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